

Phenomenological modeling of antibody response from vaccine strain composition

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The elicitation of broadly-neutralizing antibodies (bnAbs) is a major goal of vaccine design for highly mutable pathogens, such as influenza, HIV, and coronavirus. Although many rational vaccine design strategies for eliciting bnAbs have been devised, their efficacies first need to be evaluated in preclinical animal models, and, ultimately, in clinical trials. To improve the outcomes for such vaccines, it would be useful to introduce early into the development process models that can predict vaccine efficacy against arbitrary pathogen variants, particularly those not present in the vaccine formulation. Here, we develop two biologically-motivated models of antibody reactivity elicited by nanoparticle-based vaccines using only antigen amino acid sequences, parametrized with a small sample of experimental antibody binding data from influenza or SARS-CoV-2 nanoparticle vaccinations. The models are able to recapitulate the experimental data to within experimental uncertainty, are relatively insensitive to the choice of the parametrization/training set, and provide predictions about the antigenic epitopes exploited by the vaccine, which are broadly consistent with experimentally known epitopes. For the mosaic nanoparticle vaccines considered here, model results suggest indirectly that the sera obtained from vaccinated animals contain bnAbs, rather than simply different strain-specific Abs, but the difference between the two interpretations is not statistically significant, partly because of large uncertainties in the experimental data with which the model results are compared. Overall, this study suggests that simple models of vaccine efficacy parametrized with modest amounts of experimental data can be used in the vaccine development process to improve the effectiveness of designed vaccines.

Keywords: IgG, vaccination, simulation, influenza, hemagglutinin, coronavirus

I. INTRODUCTION

Infections by highly-mutable viruses cause high mortality and morbidity around the world. For example, according to the CDC, between the years 2010–2014, influenza-associated deaths in the United States alone ranged from 12,000 (winter of 2011–2012) to 56,000 (winter of 2012–2013). The currently available flu vaccines need to be reformulated yearly because the high mutation rate of the circulating flu strains continuously renders preexisting immunity obsolete. Despite the yearly adjustments, the long time lag in producing the vaccine relative to the high rate of antigenic drift can result in a low level of protective immunity (10% – 60%).¹ For example, at the time of the writing of this article, the seasonal influenza vaccine efficacy was estimated at 16%,² significantly below the approximate threshold needed for herd immunity of 50% (assuming a basic viral reproduction number R_0 ³ of 2).

The COVID pandemic of 2019 resulted in rapid development of highly effective vaccines,⁴ with effectiveness of $\geq 90\%$ against the original (Wuhan) strain.^{5,6} However, mutations in the coronavirus spike protein,⁷ compounded by high transmission rates³ and selective evolutionary pressure, exerted by vaccine-induced antibodies, caused the emergence of viral ‘escape’ variants, against which the antibodies induced by the standard prime-boost vaccination regimen had much reduced efficacy, *e.g.*, 67%–70% against the Omicron variant.⁸ Such levels of efficacy are lower than the 80% requirement for herd immunity based on a conservative R_0 estimate of 5.⁹

The situation is more dire with HIV, which causes about 680,000 yearly deaths worldwide,¹⁰ and no effective vaccine is available, despite ongoing efforts.¹¹

The above statistics demonstrate that universal, ideally permanent, vaccines for highly-mutable viral pathogens are desirable. One strategy for developing such vaccines focuses on the elicitation of broadly-neutralizing antibodies (bnAbs), *i.e.* the Abs that are able to neutralize a broad range of antigen variants,^{12,13} in contrast to strain-specific antibodies, which typically dominate the adaptive immune response.¹⁴

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Universal vaccine candidates designed to elicit broad protection typically aim to present multiple epitopes to the immune system in the form of chimeric antigens harboring a patchwork of different epitopes,¹⁵ in the form of optimized antigenic cocktails,^{16,17} or as co-display of different antigens on nanoparticles.^{18–21}

An important step toward robust rational design of such vaccines is being able to predict properties of the antibody response to vaccination. Such methods could be used in comparisons of the modeled outcomes to different vaccination cocktails or regimens, to select those that predict the highest antibody breadth or potency.

Many models exist that simulate affinity maturation (AM) inside germinal centers using concepts from biology and physics (reviewed in Ref. 22). In these models, Darwinian evolution, driven by random mutations and selection of B-cell receptors (BCRs) on the basis of their affinity for antigen, generates high-affinity antibodies. However, most of the AM models are too approximate or coarse-grained to be able to evaluate specific vaccines,^{23,24} and thus provide only general design guidelines (*e.g.*, regarding efficacy of single-antigen *vs.* cocktail vaccination,^{25,26} or regarding the effects of binding valency or antigen concentration²⁴). Although advances in computational methods and speed are enabling modeling of BCR/antigen interactions in all-atom detail,²⁷ such models are still too expensive to be used for generating meaningful statistics.

Here, we propose simple biologically-motivated phenomenological models for reactivity to antigen variants of post-vaccination sera elicited by nanoparticle vaccines with prescribed antigenic sequences. The key assumptions in the models are that (i) antibody reactivity is higher when the test strain – that against which reactivity is measured – is similar to the strains in the vaccine, and (ii) different amino acid residues make different contributions to the overall binding titer; the contributions are represented in the form of residue weights, optimized by fitting to experimental vaccination data for influenza²⁰ or SARS-CoV-2²¹. The residues with high weights can be interpreted as being involved in antibody binding, which we find to be broadly consistent with the locations of known epitopes on the influenza hemagglutinin and SARS-CoV-2 spikes.²⁸ In addition to parametrization using experimental data, the models require only the antigenic sequences, so that structural information, which could be difficult or time-consuming to obtain, is not needed. Our results suggest that modeling strategies such as the one described here can be combined with experiments to optimize vaccination cocktails for breadth or potency.

II. METHODS

IgG titer model

Sequence coordinate notation

Let $\hat{\mathbf{S}} \in M_{N_s \times N_r}$ denote a multiple alignment matrix of N_s antigen sequences (MSA), with each row $\mathbf{S}_i$ of $\hat{\mathbf{S}}$ corresponding to a unique sequence i of length N_r , possibly with gaps to facilitate the MSA. $\hat{\mathbf{S}}$ is represented by the standard 20-amino acid alphabet (A) and the gap symbol ($G \equiv -$).

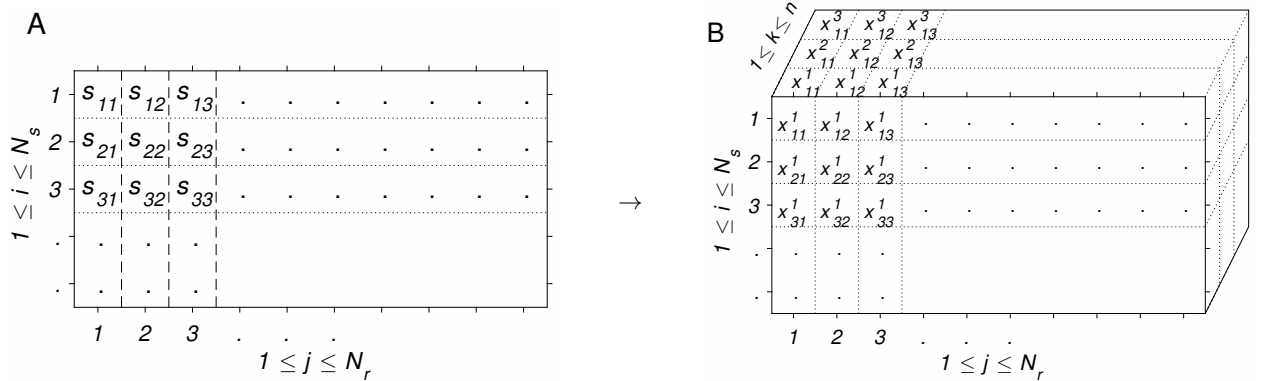


FIG. 1. Illustration of the amino acid encoding procedure.²⁹ A: Multiple alignment $\hat{\mathbf{S}}$ of N_s sequences, each having N_r residues, including gaps, with $s_{ij} \in \{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y, -\}$. B: Embedded alignment $\hat{\mathbf{X}}$; each residue type in $\hat{\mathbf{S}}$ shown in A is associated with a ($n=3$)-dimensional real-valued vector $\mathbf{x}_{ij} = \{x_{ij}^1, x_{ij}^2, x_{ij}^3\}$, which is interpreted as Cartesian coordinates of the residue.

To be able to perform standard mathematical operations on $\hat{\mathbf{S}}$, we embed $\hat{\mathbf{S}}$ in an n -dimensional (nD) vector space, as follows. To each residue s_{ij} at position j in $\mathbf{S}_i$, we assign a vector $\mathbf{x}_{ij} \in \mathbb{R}_n \equiv \mathbf{x}(s_{ij})$ (see Fig. 1). A number of such

embeddings (also called encodings³⁰) can be found in the literature,^{29–32} which have been used in machine learning models of sequence-activity relationships.^{32,33} We note that the inverse map $s(\mathbf{x})$ for arbitrary $\mathbf{x}$ is not needed in our study, because we do not create new sequences. However, such an inverse could be defined as a suitable projection onto the set of allowed residue coordinates, *e.g.*,

$$s(\mathbf{x}) = \arg \min_{s \in \text{AUG}} \|\mathbf{x} - \mathbf{x}(s)\|, \quad (1)$$

as done in Ref. 33.

Previous studies have found that the choice of embedding can have significant effects on model quality.³⁰ In the present calculations, we found that the simple 3D embedding model of Grantham²⁹, which is based on amino-acid physico-chemical properties, produces model fits that approximate IgG data to within the experimental error, with high Pearson correlation coefficients ($\simeq 0.8$ - 0.9) (see *Results*). For completeness, we also tried a higher-dimensional (5D) embedding due to Atchley *et al.*³¹, which gave slightly inferior results in terms of comparison with experiment (see Supporting Material). For simplicity, all gap positions in $\widehat{\mathbf{S}}$ are assigned the zero vector; *i.e.*, we did not attempt to optimize the MSA representation of amino acid deletions. In the remainder of this section, we use the 3D embedded coordinates x_{ij}^k with $k=1,2,3$ to represent antigen residues in the MSA.

Model 1: distance to average strain

Let $\widehat{\mathbf{X}}$ be the embedded sequence alignment as described above. We assume that a subset of v strains $V \equiv \{i_1, i_2, \dots, i_v\}$ from $\widehat{\mathbf{S}}$ has been chosen as the vaccination cocktail $\widehat{\mathbf{V}}$, and let $\overline{\mathbf{V}}$ be the average cocktail sequence. We use a simple average, but a weighted average could also be used, *e.g.*, if the strains in the cocktail had different concentrations. We also note that additional (as yet unknown) strains can readily be added to the alignment *a posteriori*, *i.e.*, after the models have been parametrized.

For any (test) sequence $\mathbf{T} \in \widehat{\mathbf{X}}$, the IgG titer against $\mathbf{T}$ elicited by the cocktail $\widehat{\mathbf{V}}$ is modeled to be

$$IgG_1^M(\widehat{\mathbf{V}}, \mathbf{T}) \equiv f(d(\overline{\mathbf{V}}, \mathbf{T}; \mathbf{W})), \quad (2)$$

where d is the weighted Euclidean distance between $\overline{\mathbf{V}}$ and $\mathbf{T}$ with weights $\mathbf{W} \in M_{1 \times N_r}$ (specified below), *i.e.*,

$$\begin{aligned} d(\overline{\mathbf{V}}, \mathbf{T}; \mathbf{W}) &= \left\{ \sum_{j=1}^{N_r} w_j^2 \|\mathbf{t}_j - \overline{\mathbf{v}}_j\| \right\}^{1/2} \\ &= \left\{ \sum_{j=1}^{N_r} w_j^2 \sum_{k=1}^n (t_j^k - \overline{v}_j^k)^2 \right\}^{1/2} \end{aligned} \quad (3)$$

and

$$f(d; \alpha, \beta) = \frac{1}{\alpha + d^\beta} \quad (4)$$

is the similarity function, with parameters α and β to be specified below. Eq. (2) assumes that IgG titers are determined by the similarity between the test strain and the *average* vaccination strain in the cocktail. The residue weights $\mathbf{W}$, as well as α and β , are optimized by fitting to experimental IgG titers, as will be described at the end of this section. The reason for the rather *ad hoc* functional form of f , is that it is one of the simplest ansätze that embodies an inverse relationship between distance and similarity, with sufficient flexibility for fitting provided through the coefficients α and β .

Model 2: average strain distance

Let $\widehat{\mathbf{V}}$ represent a vaccine cocktail, as defined above. For a sequence $\mathbf{T} \in \widehat{\mathbf{X}}$, the model IgG titer elicited by $\widehat{\mathbf{V}}$ is defined to be

$$\begin{aligned} IgG_2^M(\widehat{\mathbf{V}}, \mathbf{T}) &\equiv f \left\{ \frac{1}{v} \sum_{\mathbf{V} \in \widehat{\mathbf{V}}} d(\mathbf{V}, \mathbf{T}; \mathbf{W}) \right\} \\ &= f \left\{ \bar{d}(\widehat{\mathbf{V}}, \mathbf{T}) \right\}, \end{aligned} \quad (5)$$

where $\bar{d}(\hat{\mathbf{V}}, \mathbf{T})$ is the average distance between the vaccine cocktail $\hat{\mathbf{V}}$ and strain $\mathbf{T}$. Eq. (5) assumes that IgG titers are determined by the average distance between the test strain and each strain in the vaccine cocktail. For simplicity, we use the same functional form for the similarity function (Eq. (4)). As for Model 1, the weights and similarity function coefficients are optimized by fitting to experimental IgG measurements.

Intuitively, we would expect Model 1 to be more predictive if the elicited immunity is composed of cross-reactive antibodies, and Model 2 to be more predictive if the elicited immunity is composed of multiple strain-specific antibodies. More specifically, if a particular B-cell receptor (BCR) has significant affinities for multiple strains in the cocktail *via* the same general epitope or region on the antigen, we assume that the overall selection pressure acting on the corresponding B-cell during affinity maturation (AM) will be the average of the individual (strain-specific) selection pressures. Because the net result of AM is to improve the binding affinity of BCRs to their cognate antigen(s) by improving the structural complementarity of the BCR paratope to the epitope *via* interfacial residue composition matching²⁷, we assume that the *effective* evolutionary driving force is directed toward an “average” epitope. As the structures of the antigens are generally unknown, we model effects of the average structure through the average sequence, denoted above by $\bar{\mathbf{V}}$. We note that such an AM process is similar to the idea of “frustration”^{23,34}, whereby different co-administered antigens exert conflicting selection pressures on BCR evolution, resulting in some *averaged* or *compromise* driving force.

We hypothesize that maturing B-cells that undergo the above type of selection would develop affinity to an “average” epitope structure; thus, the affinity of the elicited response to any antigen strain $\mathbf{T}$ is assumed to be determined by the distance between $\mathbf{T}$ and the average vaccine strain $\bar{\mathbf{V}}$, as in Model 1. If most of the Abs elicited by vaccination arise from such BCRs, then we would expect Model 1 to be superior to Model 2 in recapitulating the experimental data.^{20,21} We would also expect such Abs to be cross-reactive in the antigenic “subspace” spanned by the antigens that are present in the vaccine.

In an alternative maturation scenario, any particular immature BCR would have significant affinity only to a single antigen in the cocktail. In this case, the selection forces driving the AM of the BCR will be dominated by that antigen. If most of the Abs elicited by vaccination evolve in this way, the Ab repertoire will be dominated by an ensemble of strain-specific Abs; in this case, we assume that the affinity of the elicited response to an antigen strain $\mathbf{T}$ will be a function of the average distance from $\mathbf{T}$ to the constituent vaccine strains, as in Eq. (5), and we would expect Model 2 to yield a superior recapitulation of the experimental affinity data.

We caution, however, that conceptual differences in the interpretations of Models 1 & 2 are limited in practice by the fact that we use a single set of residue weights $\mathbf{W}$ in computing each distance in Eq. (5). Although the model in Eq. (5) is trivially generalizable for use with multiple weight vectors, which could be interpreted as, *e.g.*, quantifying the contributions of distinct antigen epitopes, we are presently limited by the small size of the experimental data sets to meaningfully explore more complex models.

Model parameter optimization

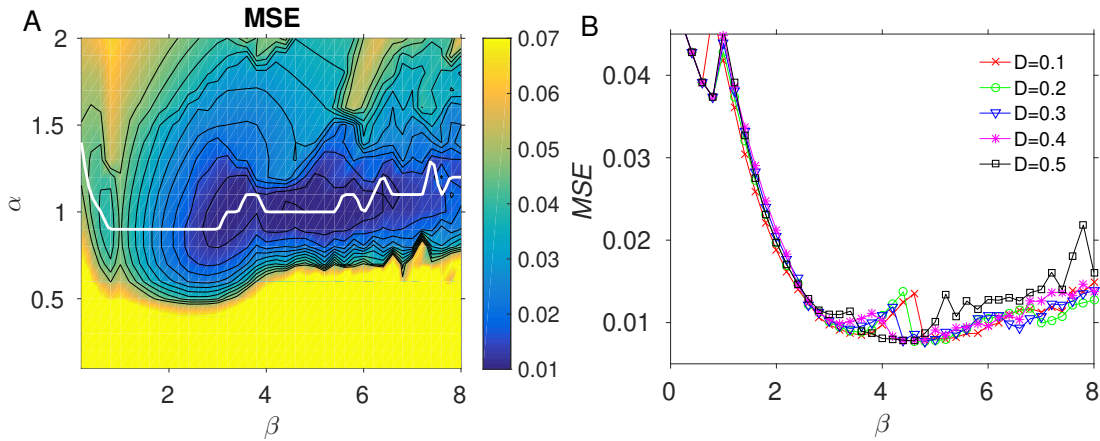


FIG. 2. Results of parameter optimization (α, β) for Model 1 applied to influenza data²⁰ using a diffusion constant $D=0.4$. A: 2D scan of the (α, β) landscape; the white line corresponds to a cubic spline curve through the minimum values of the mean squared error (MSE) over the range of α at each β ; B: MSE corresponding to the white line in A plotted in 1D for several values of the diffusion constant D .

As described in the previous subsection, fitting the two models to experimental data involves optimization of the unspecified parameters α , β and the residue weights $\mathbf{W}$. Because the 2D parameter landscape of (α, β) was discovered to have multiple local minima, we performed a 2D scan of the parameter space to find optimal values (see Fig. 2A). However, optimization of residue weights $\mathbf{W}$ in this fashion would be infeasible, because of the large number of residues (~ 550 for the influenza hemagglutinin MSA). Therefore, the weights were optimized using steepest gradient descent, as follows.

Let $IgG^E(\hat{\mathbf{V}}, \mathbf{T})$ denote the experimental ELISA titer against test antigen $\mathbf{T}$ obtained using sera from animals vaccinated with cocktail $\hat{\mathbf{V}}$,^{20,21} and let $\delta IgG^E(\hat{\mathbf{V}}, \mathbf{T})$ denote the corresponding experimental uncertainty. The weighted squared error (WSE) of the antigen test set $\hat{\mathbf{T}}$ is defined as

$$\epsilon_{\hat{\mathbf{V}}, \hat{\mathbf{T}}}^2 = \sum_{\mathbf{T} \in \hat{\mathbf{T}}} \left[\frac{IgG^M(\hat{\mathbf{V}}, \mathbf{T}) - IgG^E(\hat{\mathbf{V}}, \mathbf{T})}{\delta IgG^E(\hat{\mathbf{V}}, \mathbf{T})} \right]^2, \quad (6)$$

where the dependence on $\mathbf{W}$ enters *via* Eqs. (2) and (5), and the subscripts on ϵ^2 make explicit the dependence on $\hat{\mathbf{V}}$ and $\hat{\mathbf{T}}$, which will be omitted below for brevity. Starting from uniform initial weights $\mathbf{W}=0.1$, and values for (α, β) corresponding to the current position in the 2D scan (discussed further below), the weights are updated according to

$$\begin{aligned} \mathbf{W}^* &= \mathbf{W}^n - \tau \nabla_{\mathbf{W}} \epsilon^2 + D \nabla^2 \mathbf{W}^n, \\ \mathbf{W}^{n+1} &= \max\{\mathbf{W}^*, 0\}, \end{aligned} \quad (7)$$

where τ is the steepest descent step, $\nabla_{\mathbf{W}}(\cdot)$ denotes gradients *w.r.t.* $\mathbf{W}$, $\nabla^2 \mathbf{W}^n$ is the discrete analog of the 1D Laplacian of $\mathbf{W}^n$ *w.r.t.* residue position in the MSA, and D is an empirically selected diffusion constant. In the above, the diffusion term is added as a type of regularization, to enforce smoothness of $\mathbf{W}$ as a function of residue position. This approach reduces overfitting to experimental data, and also has a physical motivation: we can interpret the residues with high weights as those that contribute to antibody binding. The diffusion regularization term ensures that residues that are close to each other do not make vastly different contributions to binding. The sensitivity of the fits to the value of D is relatively minor, as will be shown below. The discrete Laplacian in our computer implementation of Eq. (7) is

$$\nabla^2 w_i^n \equiv \begin{cases} w_{i-1}^n - 2w_i^n + w_{i+1}^n, & i = 2, \dots, N_r - 1 \\ 0, & i = 1, N_r, \end{cases} \quad (8)$$

which is equivalent to the finite difference of the second derivative computed to second order. The last line of Eq. (7) ensures that $\mathbf{W}$ are nonnegative at every step.

The expressions for the gradient $\nabla_{\mathbf{W}} \epsilon^2$ are obtained from Eqs. (2)-(5). For Model 1,

$$\begin{aligned} \frac{\partial \epsilon^2}{\partial w_k} &= \sum_{\mathbf{T} \in \hat{\mathbf{T}}} \left[\frac{IgG_1^M(\hat{\mathbf{V}}, \mathbf{T}) - IgG^E(\hat{\mathbf{V}}, \mathbf{T})}{\delta IgG^E(\hat{\mathbf{V}}, \mathbf{T})^2} \right] \\ &\quad \times 4f'(d^2)w_k \|\bar{\mathbf{v}}_k - \mathbf{t}_k\|, \end{aligned} \quad (9)$$

where we wrote f as a function of d^2 rather than d to avoid the singularity of $\nabla_{\mathbf{W}} d$ at $d=0$.

For Model 2,

$$\begin{aligned} \frac{\partial \epsilon^2}{\partial w_k} &= \sum_{\mathbf{T} \in \hat{\mathbf{T}}} \left[\frac{IgG_2^M(\hat{\mathbf{V}}, \mathbf{T}) - IgG^E(\hat{\mathbf{V}}, \mathbf{T})}{\delta IgG^E(\hat{\mathbf{V}}, \mathbf{T})^2} \right] \\ &\quad \times 2f'(\bar{d}) \frac{1}{v} \sum_{\mathbf{V} \in \hat{\mathbf{V}}} \frac{w_k \|\mathbf{v}_k - \mathbf{t}_k\|}{d(\mathbf{V}, \mathbf{T}; \mathbf{W})}. \end{aligned} \quad (10)$$

Note that if $\hat{\mathbf{V}}$ and $\hat{\mathbf{T}}$ have strains in common, $d(\mathbf{V}, \mathbf{T}; \mathbf{W})$ will be 0 for those strain pairs, and Eq. (10) will be singular; in such cases, we take the contribution to the gradient to be zero.

The above error minimization equations are written under the assumption of a particular vaccination cocktail $\hat{\mathbf{V}}$. However, we optimized a single set of weights for all of the vaccination cocktails in Ref. 21 or Ref. 20 simultaneously. In that case, the total squared error (and its gradients) are simply the sums of those corresponding to the individual vaccinations.

The results of optimizing the similarity function parameters (α, β) in Eq. (4) are illustrated in Fig. 2 for fitting Model 1 to influenza data.²⁰ Figure 2A shows a 2D discrete scan of the parameter space $(\alpha, \beta) \in \{0, 0.1, 0.2, \dots, 2\} \times \{0, 0.2, 0.4, \dots, 8\}$. For each grid point, the weights $\mathbf{W}$ were reoptimized according to Eq. (7) starting from $\mathbf{W}^0=0.1$, and iterating for $n_{max}=3000$ steps with $\tau=0.03$, diffusion constant $D=0.4$, and the fit mean squared error computed from Eq. (6) and averaged. The scan shows a broad, irregularly-shaped minimum for $\alpha \simeq 1, \beta \gtrsim 3$; however, for $\beta \gtrsim 5$, the optimization landscape shows oscillatory behavior, suggesting that the model becomes sensitive to parameters, or that the gradient descent step could be too large in this region. However, the 1D plot through the minimum at each β value (Fig. 2B) shows that the global minimum lies within the region $\beta \in (3, 5)$, and suggests that further parameter space optimization or exploration are not necessary. We repeated the optimization for multiple values of D (Fig. 2B), and observed that the error minimum and location are not very sensitive to the choice of D ; however, for $D \gtrsim 0.5$, we observed increased ruggedness, as can be seen for $D=0.5$ (black line in Fig. 2B). For the model calculations in *Results*, we used the value $D=0.49$, which results in a relatively smooth distribution of the residue weights, as shown in the Results (*e.g.* Fig. 3D or Fig. S4). We also noticed that the weights optimization could be performed with fewer iterations and at higher τ , with values up to 0.04, if the optimized weights were taken to be those that minimize the error over the *entire* optimization history (not just the final $\mathbf{W}^n$ in Eq. (7)), *i.e.*,

$$\mathbf{W}_{opt} = \arg \min_{0 \leq n \leq n_{max}} \epsilon^2(\mathbf{W}^n). \quad (11)$$

Parameter optimization for Model 2 and for both models applied to coronavirus data was very similar. The number of weights $\mathbf{W}$ to optimize was equal to the number of residue positions in the underlying MSA, which are 549 and 273 for influenza and coronavirus respectively, subject to the constraints of nonnegativity and diffusional regularization described above.

The parameters used to obtain the data in *Results* are given in Table III. Computer calculations were performed on a x86.64 computers running Linux using Matlab³⁵ and GNU Octave³⁶, which is an open source program syntactically compatible with Matlab. Molecular structures for the visualization of weights were rendered using Visual Molecular Dynamics.³⁷

III. RESULTS

First, we show that both models described in *Methods* are able to reproduce measurements of post-vaccination animal sera binding to panels of influenza²⁰ and SARS-Cov-2²¹ antigens to within experimental error. The virus strains and vaccine compositions are given in Tables IV and V, for flu and coronavirus, respectively. Alignments of the antigen sequences and pairwise antigen similarity matrices are provided in supporting text, and in Figs. S9 and S10. Our focus is on mosaic nanoparticles (nps) – those that display the same protein antigen but from different strains – because such vaccines appear to preferentially elicit bnAbs;^{18,19,21,38} see also *Discussion*. Fits to the titers for the mixtures of homogeneous influenza nps (admix) are shown in the Supporting Material. Model performance is summarized in Tables I and II for influenza and coronavirus, respectively.

A. Influenza Results

In Figs. 3 and 4 we show the results of Model 1 (distance-to-average) and Model 2 (average-distance), respectively, parametrized and tested using the the influenza vaccination data. In each figure, panels A and B show the optimized fits of the corresponding model to the IgG titers measured as area-under-curve (AUC) 21 days after vaccinating mice with mosaic nps (also shown in Fig. 5 of Ref. 20).

Model 1 shows a slightly superior fit to the average IgG titers than Model 2 (rmse=0.09, $c_P=0.93$ *vs.* rmse=0.13, $c_P=0.88$), where rmse is the root-mean-squared error, and c_P is the Pearson correlation coefficient. However, the rmse of the corresponding experimental IgG error is 0.23, which is higher than that for either fit. The experimental error bars correspond to the standard deviations (σ) of the titer measurements; because there were four measurements of each titer, they also correspond to 2σ of the mean IgG, or the 95% confidence range. Panels A & B in Figs. 3 and 4 show that, for both cases, all of the modeled titers for both models are within the corresponding error range. Thus, the better fit of Model 1 to the experimental data is not statistically significant.

To evaluate the predictive ability of the models, we performed the fitting procedure using three strategies. First, we carried out the fitting (or training) procedure with half of the strains (five), and used the remaining five strains to evaluate the fits. To eliminate the training/test set selection bias, we considered all $\binom{10}{5}=252$ possible partitions. Panels C of Figs. 3 and 4 show the scatter plots of the Pearson and Spearman (c_S) correlations computed over the training *vs.* test sets, and the correlation statistics are summarized in Table I (section I). In each model, the

TABLE I. Summary of model performance using influenza vaccination data.

Model-Enc-nptype	$c_P^{trn}(\text{m}\pm\text{sd})$	$c_P^{tst}(\text{m}\pm\text{sd})$	$c_S^{trn}(\text{m}\pm\text{sd})$	$c_S^{tst}(\text{m}\pm\text{sd})$	$p_{val}^{c_P, tst}(\text{min}/\text{max}/\text{avg})$	$c_P^{hld}(p_{val})$	$c_S^{hld}(p_{val})$
<i>I. Train/test split by antigen</i>							
Model 1-Gr-mosaic	0.91 $\pm$ 0.03	0.84 $\pm$ 0.05	0.92 $\pm$ 0.03	0.85 $\pm$ 0.05	2.9e-09/3.1e-03/4.8e-05	N/A	N/A
Model 2-Gr-mosaic	0.87 $\pm$ 0.03	0.78 $\pm$ 0.08	0.88 $\pm$ 0.03	0.79 $\pm$ 0.08	3.6e-08/4.0e-02/8.8e-04	N/A	N/A
Model 1-At-mosaic	0.91 $\pm$ 0.03	0.74 $\pm$ 0.13	0.91 $\pm$ 0.03	0.78 $\pm$ 0.09	4.4e-09/8.4e-02/6.7e-03	N/A	N/A
Model 1-Gr-admix	0.77 $\pm$ 0.06	0.66 $\pm$ 0.11	0.73 $\pm$ 0.07	0.65 $\pm$ 0.15	3.1e-06/1.3e-01/9.3e-03	N/A	N/A
Model 2-Gr-admix	0.72 $\pm$ 0.06	0.65 $\pm$ 0.08	0.76 $\pm$ 0.05	0.66 $\pm$ 0.08	1.7e-05/7.2e-02/5.7e-03	N/A	N/A
<i>II. Train/test split by vaccine</i>							
Model 1-Gr-mosaic	0.92 $\pm$ 0.04	0.84 $\pm$ 0.10	0.92 $\pm$ 0.03	0.86 $\pm$ 0.07	3.4e-09/2.0e-03/3.4e-04	N/A	N/A
Model 2-Gr-mosaic	0.89 $\pm$ 0.06	0.80 $\pm$ 0.13	0.88 $\pm$ 0.06	0.81 $\pm$ 0.11	8.1e-09/5.9e-03/1.1e-03	N/A	N/A
Model 1-At-mosaic	0.93 $\pm$ 0.03	0.77 $\pm$ 0.11	0.90 $\pm$ 0.03	0.78 $\pm$ 0.11	9.7e-07/3.6e-03/1.0e-03	N/A	N/A
Model 1-Gr-admix	0.85 $\pm$ 0.20	0.71 $\pm$ 0.19	0.78 $\pm$ 0.19	0.70 $\pm$ 0.19	1.8e-10/9.6e-02/1.6e-02	N/A	N/A
Model 2-Gr-admix	0.88 $\pm$ 0.16	0.67 $\pm$ 0.21	0.80 $\pm$ 0.15	0.66 $\pm$ 0.10	2.8e-07/1.5e-01/2.8e-02	N/A	N/A
<i>III. With a ‘holdout’ set (V8)</i>							
Model 1-Gr-mosaic	0.92 $\pm$ 0.02	0.74 $\pm$ 0.07	0.92 $\pm$ 0.04	0.70 $\pm$ 0.09	1.6e-03/1.2e-01/1.9e-02	0.79(6.9e-03)	0.73(2.1e-02)
Model 2-Gr-mosaic	0.93 $\pm$ 0.02	0.87 $\pm$ 0.04	0.88 $\pm$ 0.04	0.82 $\pm$ 0.06	3.7e-05/1.2e-02/1.6e-03	0.89(4.9e-04)	0.82(6.8e-03)
Model 1-At-mosaic	0.92 $\pm$ 0.02	0.71 $\pm$ 0.11	0.88 $\pm$ 0.05	0.69 $\pm$ 0.13	1.4e-04/7.7e-01/3.7e-02	0.77(9.4e-03)	0.71(2.8e-02)
Model 1-Gr-admix	0.90 $\pm$ 0.04	0.62 $\pm$ 0.09	0.66 $\pm$ 0.12	0.53 $\pm$ 0.14	9.8e-03/3.4e-01/7.0e-02	0.70(2.5e-02)	0.63(5.3e-02)
Model 2-Gr-admix	0.96 $\pm$ 0.01	0.81 $\pm$ 0.05	0.83 $\pm$ 0.06	0.68 $\pm$ 0.07	1.1e-03/1.8e-01/7.1e-03	0.85(1.8e-03)	0.70(2.4e-02)

Abbreviations **Gr** and **At** refer to the Grantham²⁹ and Atchley³¹ encodings, respectively; **nptype** refers to the nanoparticle type which can be mosaic or admix;²⁰ subscripts *trn* and *tst* refer to training and test sets, respectively. P-values (p_{val}) are reported for the Pearson correlation coefficient computed over the test set as triplets, indicating the minimum, maximum, and average values. For testing on the ‘holdout’ set, the performance of the models is indicated by Pearson (c_P^{hld}) and Spearman (c_S^{hld}) correlation coefficients with the corresponding P-values in parentheses; N/A indicates that a holdout set was not chosen.

TABLE II. Summary of model performance using coronavirus vaccination data.

Model-Enc	$c_P^{trn}(\text{m}\pm\text{sd})$	$c_P^{tst}(\text{m}\pm\text{sd})$	$c_S^{trn}(\text{m}\pm\text{sd})$	$c_S^{tst}(\text{m}\pm\text{sd})$	$p_{val}^{c_P, tst}(\text{min}/\text{max}/\text{avg})$	$c_P^{hld}(p_{val})$	$c_S^{hld}(p_{val})$
<i>I. Train/test split by antigen</i>							
Model 1-Gr	0.86 $\pm$ 0.04	0.59 $\pm$ 0.13	0.84 $\pm$ 0.06	0.56 $\pm$ 0.13	9.6e-05/2.5e-01/3.9e-02	N/A	N/A
Model 2-Gr	0.80 $\pm$ 0.09	0.53 $\pm$ 0.13	0.78 $\pm$ 0.08	0.54 $\pm$ 0.14	3.5e-05/3.4e-01/6.2e-02	N/A	N/A
<i>II. Train/test split by vaccine</i>							
Model 1-Gr	0.89 $\pm$ 0.03	0.65 $\pm$ 0.16	0.84 $\pm$ 0.12	0.64 $\pm$ 0.16	1.8e-06/3.8e-02/1.5e-02	N/A	N/A
Model 2-Gr	0.87 $\pm$ 0.05	0.55 $\pm$ 0.14	0.83 $\pm$ 0.08	0.59 $\pm$ 0.11	1.3e-04/1.2e-01/3.9e-02	N/A	N/A
<i>III. With a ‘holdout’ set (V8)</i>							
Model 1-Gr	0.87 $\pm$ 0.04	0.68 $\pm$ 0.12	0.86 $\pm$ 0.04	0.63 $\pm$ 0.15	9.1e-04/6.8e-01/6.5e-02	0.80(9.0e-03)	0.83(8.3e-03)
Model 2-Gr	0.83 $\pm$ 0.07	0.69 $\pm$ 0.08	0.85 $\pm$ 0.05	0.63 $\pm$ 0.09	2.5e-03/1.7e-01/4.7e-02	0.73(2.6e-02)	0.60(9.7e-02)

The quantities listed are the same as those described for influenza in Table I.

correlations for the training set ($c_P \sim 0.91$ and ~ 0.87 for Models 1 & 2, respectively) are generally higher than those for the testing set ($c_P \sim 0.84$ and ~ 0.78 , resp.). This is expected, because the former, but not the latter, set was used in the optimization/fitting. However, since the correlations are rather high even for the test set, it appears that overfitting to the training data is not problematic. Further, comparing the distributions of the correlation coefficients for Models 1& 2 in terms of the σ , the differences in fit quality are not significant even to within 1σ or 68% confidence.

Second, we performed the fitting procedure using data from two of the four vaccine titers, corresponding to $\binom{4}{2}=6$ possible partitions. The results are summarized in Table I (section II). They are similar to those for the previous case, with the main difference being higher statistical errors in the correlation coefficients, as can also be seen from scatter plots of the Pearson and Spearman correlations in Fig. S5.

In the third, and final, check, we removed the titers corresponding to vaccine V8 from the experimental data set, and trained the models using 70% of the remaining data (*i.e.*, 7 out of 10 strains). Taking 70%, rather than 50%, of the remaining data implies that the fraction of the training set relative to the size of the entire experimental sample is $0.7 \times 0.75 = 0.525$, which is roughly consistent with the 50% used in the first two cases above. The V8 vaccine titers

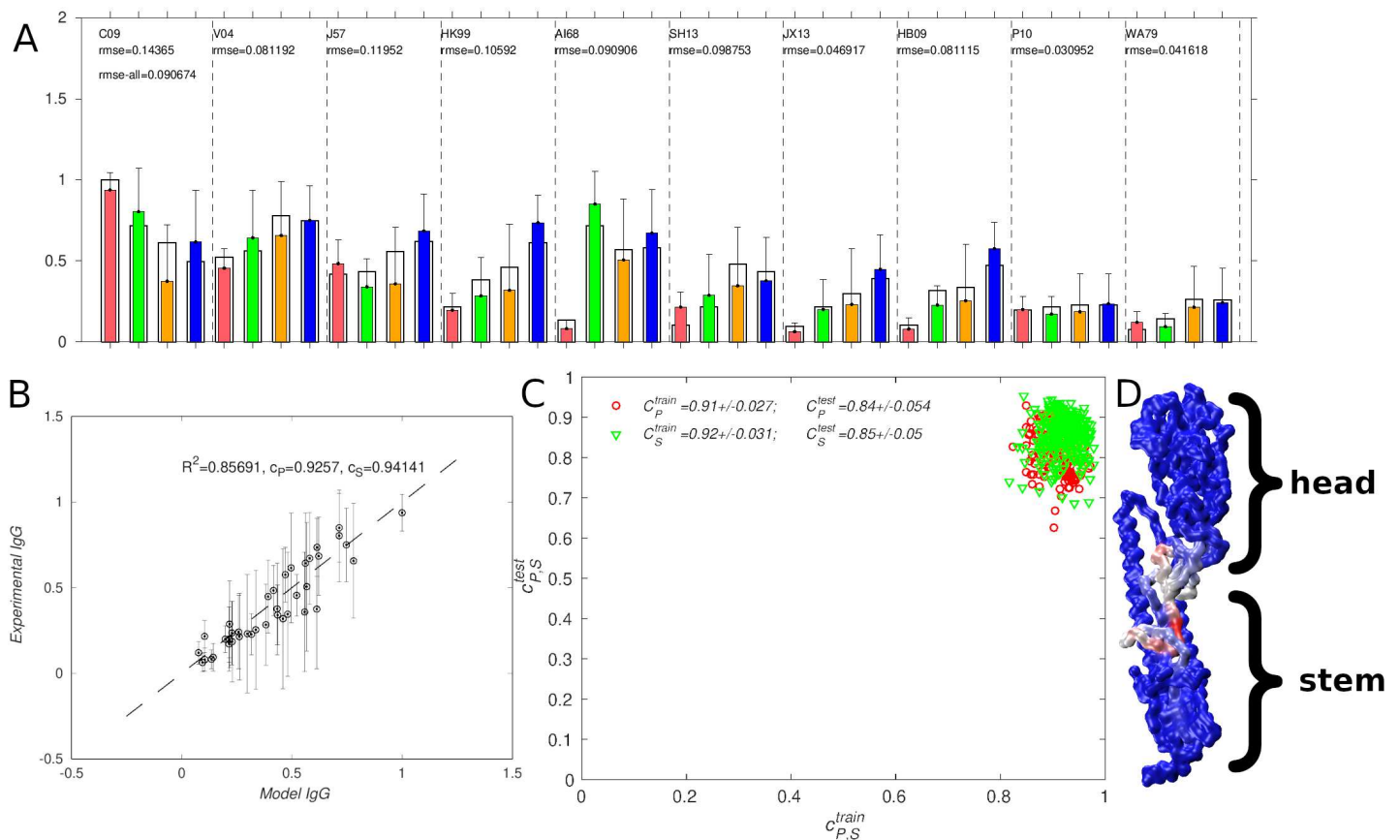


FIG. 3. Comparison of Model 1 results with the experimental IgG titers for influenza.²⁰ A & B: best fit to experiment; in A, the colors correspond to experimental data for vaccines in Table IV (red $\blacksquare$ V1, green $\blacksquare$ V2, orange $\blacksquare$ V4, blue $\blacksquare$ V8), and the black outer bars are model values; C: comparison of 252 possible fits in which 5 strains were used for fitting (training) and 5 for testing (red $\circ$: C_P , green ∇ : C_S) ; D: influenza hemagglutinin (PDB ID: 3LZG³⁹) monomer colored by model weights using the color map blue-gray-red (corresponding to low-medium-high).

were chosen as a ‘holdout’ set because the V8 vaccine has some component strains that are not present in each of the remaining vaccines (Table IV), making it a nontrivial test case of a possible novel vaccine. To generate a single computational prediction for each of the measurements in the holdout set without model selection bias, the computed titers from all of the models corresponding to the different train/test partitions were averaged prior to computing correlations in Table I (holdout columns). Overall, the correlation coefficients are comparable to those from the previous two tests, the main difference being somewhat higher p-values, indicating lower statistical significance (see column 6 in Table I). On average, the p-values correspond to greater than 95% significance, but two models (Model 1 with the Atchley encoding, and Model 2 applied to the admix data set) produce instances of particularly low statistical significance (p-values of 0.77 and 0.34, respectively). The low significance values were present despite the high Pearson correlation coefficients for the training set (≥ 0.9); A possible explanation is that the models were overfit to the training data. Model tests on the holdout set generally produced somewhat lower correlation coefficients, in the range 0.63 – 0.89 with p-values ≤ 0.053 (see columns 7 & 8 in Table I), still indicating satisfactory performance. Scatter plots corresponding to Table I section III are provided in Fig. S7. Overall, from Figs. 3 and 4 panels A-C and Table I, we can conclude that both models appear to recapitulate the IgG titer data from influenza nanoparticle vaccinations of Cohen *et al.*²⁰ with reasonable accuracy; the models give better results for the mosaic vaccine dataset, and with the Grantham²⁹ encoding.

As described in *Methods*, a feature of the present modeling is that the residue weights $\mathbf{W}$ are interpretable as reflecting contributions to antibody binding. For example, we hypothesize that residues with high weights are those that form the antibody/antigen binding interface. If true, the present models could aid in the identification of antigenic epitopes. To investigate this possibility, in panel D of Figs. 3 and 4 we show the structure of an H1 hemagglutinin (HA) monomer³⁹ colored by residue weights; the weights are also plotted as a function of residue position in Fig. S4A. In the figures, the highest weights correspond to the upper portion of the HA stem, which is known to be targeted by

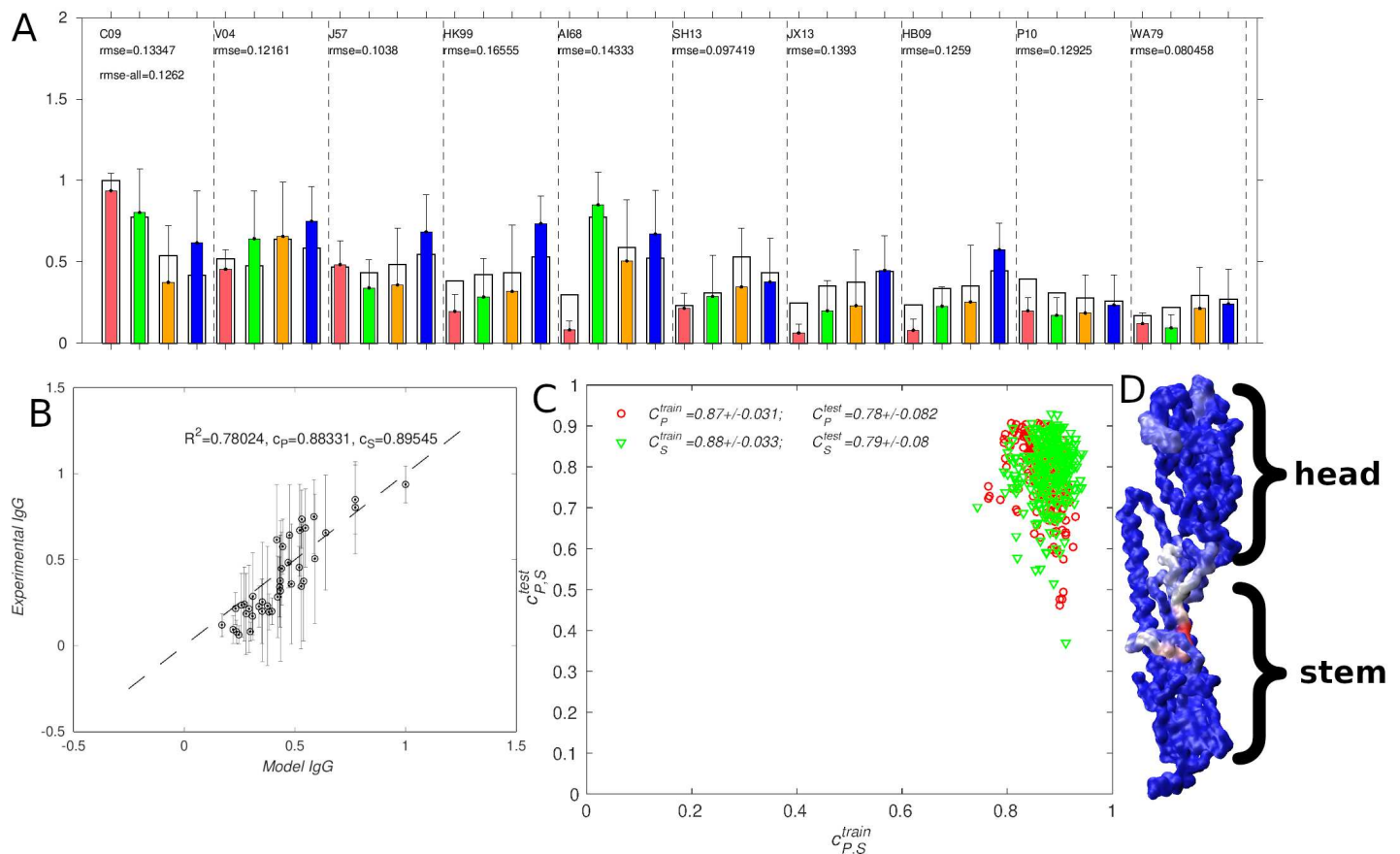


FIG. 4. Comparison of Model 2 results with the experimental IgG titers for influenza²⁰. The legends for the panels A-D are the same as those in Fig. 3.

several bnAbs.¹² In addition, the figure shows a residue patch along the top inner portion of the HA head, which is in the vicinity of the occluded epitope found by Watanabe *et al.*⁴⁰ These observations are broadly consistent with our hypothesis. However, we note that the portion of the HA stem corresponding to the HA2 subunit does not contribute to binding in the above models, as the residue weights are zero in that region (see Fig. S4A). Thus, the models may be too simple or approximate to identify all relevant binding epitopes. We note that, to actually map the epitopes to which the vaccination sera react, would require antibody competition binding experiments.

B. Coronavirus Results

Figures 5 and 6 show the results of Model 1 and Model 2, respectively, parametrized and tested using coronavirus vaccination IgG titers measured as AUC 14 days after vaccination with mosaic nps.²¹ The main difference between these data and the influenza data is that model results are of somewhat lower quality, which can be seen in the model fits (panels A & B in Figs. 5 and 6). As is the case for the influenza data, Model 1 fit follows the mean IgG titers slightly better than Model 2 (rmse=0.19, $c_P=0.81$, $c_S=0.80$ *vs.* rmse=0.23, $c_P=0.75$, $c_S=0.78$). However, the difference between the model predictions is not statistically significant. Qualitatively, this can be seen from the fact that most of the modeled values in Figs. 5 and 6 A & B are within the experimental error bar. The experimental rmse in the titers, with ten replicates per value, is 0.36;²¹ the 95% confidence limit for the average value would correspond to a reduction of the error bar by a factor of only $2/\sqrt{10} \simeq 0.63$, giving 0.23, which is comparable to the rmse for both models. Quantitatively, we evaluate the performance of the models using three strategies, as described for influenza. First, we compare the distributions of the correlations obtained when the titer data are split into a training set with 5 AG strains, and the remaining 4 strains are used for testing. There are $\binom{9}{5}=126$ such set pairs, corresponding to scatter plots in Figs. 5 and 6 panels C.

As before, the correlations for the training sets ($c_P \sim 0.86$ and ~ 0.80 for Models 1 & 2, respectively) are higher

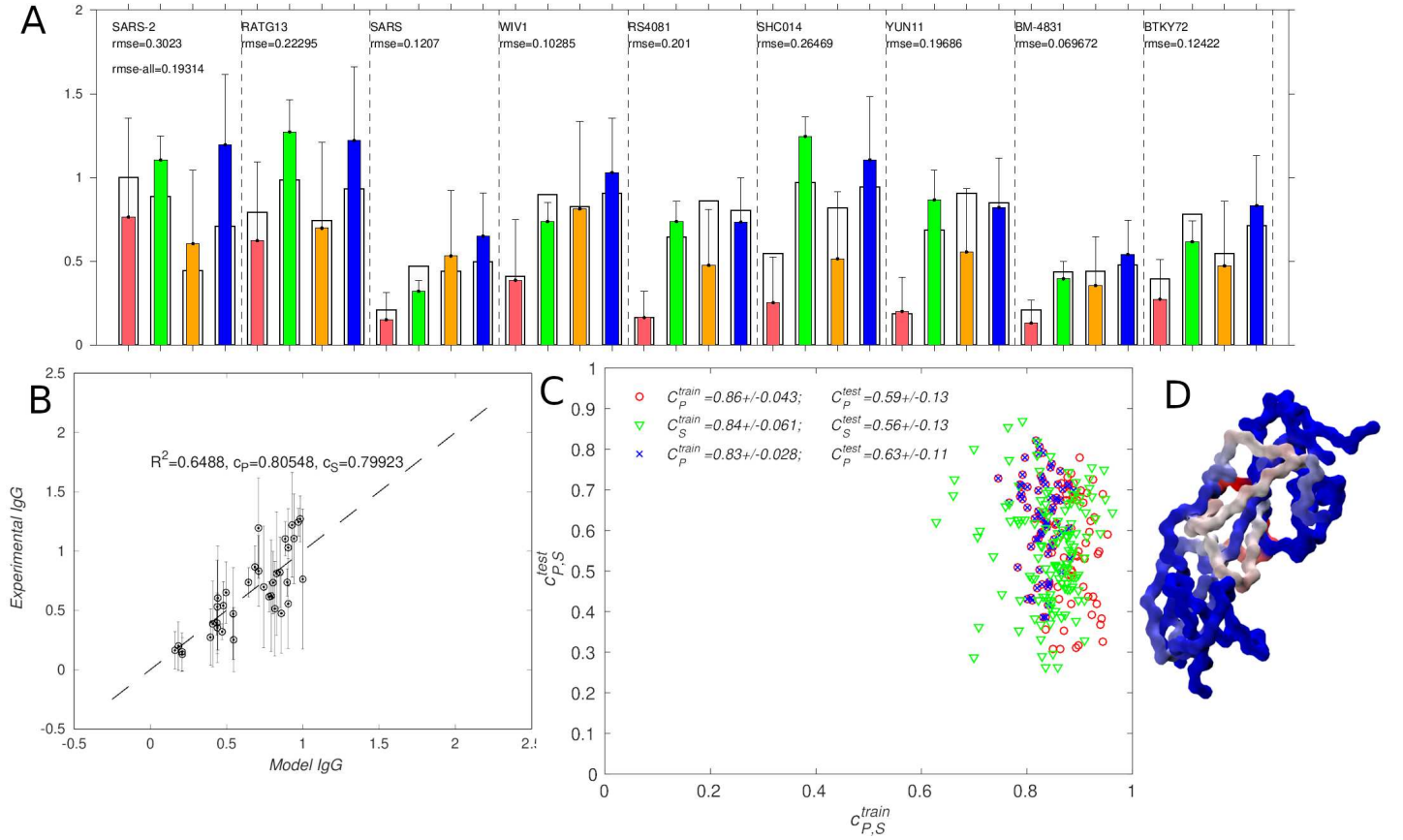


FIG. 5. Comparison of Model 1 results with the experimental IgG titers for coronavirus²¹. A & B: best fit to experiment; in A, the colors correspond to experimental data for vaccines in Table V (red $\blacksquare$ V1, green $\blacksquare$ V4A, orange $\blacksquare$ V4B, blue $\blacksquare$ V8), and the black outer bars are model values; C: comparison of 126 possible fits in which 5 strains were used for fitting (training) and 4 for testing (red $\circ$: C_P , green ∇ : C_S , blue $\times$: C_P for training sets that include SARS-2); D: SARS-CoV-2 receptor binding domain (RBD), colored by model weights using the color map blue-gray-red (corresponding to low-medium-high); PDB ID: 6VXX⁴¹.

than those for the testing set ($c_P \sim 0.59$ and ~ 0.53 , resp.). These differences are somewhat higher than those for the influenza data, suggesting that there is more overfitting to the training set for the coronavirus data. Comparison of the distributions of the correlations in terms of the σ , shows that the differences in prediction quality are not significant to within 1σ . Thus, both models provide similar fits to the data. Further, although, on the average, the p-value for the Pearson correlation coefficients are 0.039 and 0.062 for Models 1 & 2, respectively, the maximum values are 0.25 and 0.34, respectively, indicating that some correlation values are not very significant (see Table II section I).

Next, we split the experimental data along vaccine groups, choosing two groups for training and two for testing, for a total of six possible splits, as was done for influenza. The results are similar to the first test above ($c_P^{test} \sim 0.65$ and ~ 0.55 for Models 1 & 2, respectively), and the maximum p-values are somewhat lower for the two models (0.038 and 0.12, respectively; Table II section II). The correlation scatter plots are shown in Fig. S6.

Finally, we evaluate the models on three-way data splits, in which all IgG titers corresponding to vaccine V8 are set aside as a holdout set, and the two thirds of the remaining data (six out of nine antigen strains) are used for training. Using two thirds corresponds to 50% of the total data, consistent with the the above two checks, and with the testing procedure for influenza. Applying the models to the holdout set and averaging the computed titers over the models, as done above for influenza, results in the Pearson correlations of 0.8 and 0.73 for Model 1 and 2, respectively, with p-values ≤ 0.026 (Table II section III). This result appears acceptable, especially given the small quantity of training data. The correlation scatter plots for this case are shown in Fig. S8.

It is noteworthy that the largest discrepancy between the model and experimental results corresponds to the IgG titers obtained after vaccination with only the SARS-Cov-2 receptor-binding domain (RBD) antigen, measured against the same antigen (Figs. 5 and 6 panel A, first bar from the left). Because the test strain and the vaccine strain are identical, one would expect to see strong binding between the vaccination sera and the antigen. However, some serum samples inexplicably showed no detectable binding in the experiment,²¹ which caused the high error bar, and the low

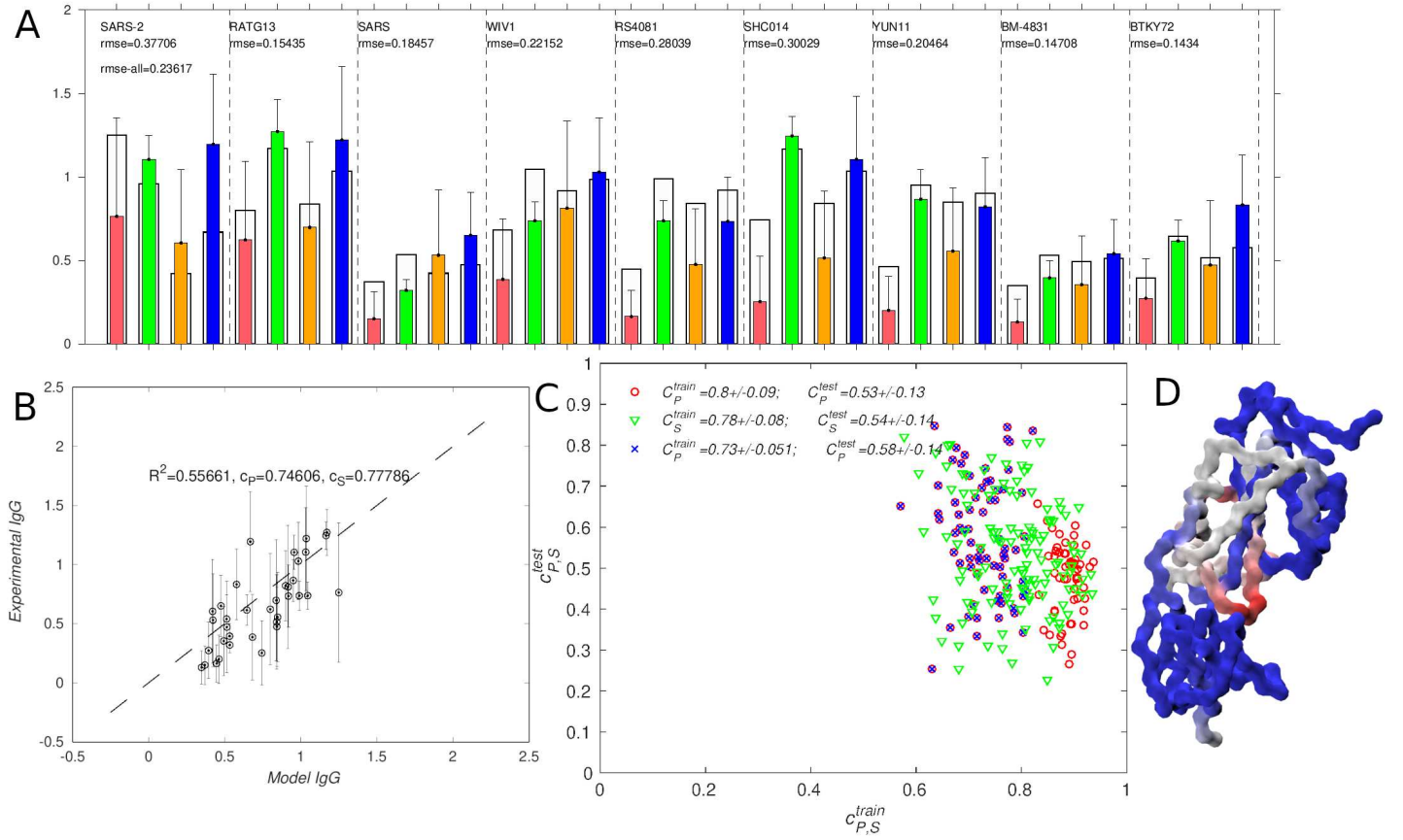


FIG. 6. Comparison of Model 2 results with the experimental IgG titers for coronavirus²¹. The legends for the panels A-D are the same as those in Fig. 5.

average titer, compared with vaccines V4A (green bars) and V8 (blue bars) (see Table V & Figs. 5 and 6).

This result seems at odds with the main assumption made in our models, *i.e.* that sera reactivity increases with the similarity of the test strain to the vaccination strain(s) (Eqs. (2) and (5)), and, in part, explains the lower fit and prediction quality. Indeed, Figs. 5 and 6 panels A show that the rmse between the experiment and model is highest for the SARS-2 test strain.

Other strains for which the discrepancies between the model and experimental IgG values are large, though not as large as for SARS-2 described above, are RS4081, SHC014, and YUN11, especially when measured *w.r.t.* sera of vaccine V4B (orange bars in Figs. 5 and 6). However, none of those strains are components of V4B (see Table V). Therefore the titer differences are, in principle, justifiable as modeling errors due to strain mismatch, as well as due to non-binding experimental samples. In contrast, for the SARS-2 case, the match between the vaccine and test strains is 100%, which is why a low titer appears unphysical. Although it is tempting to consider the non-binding samples as being erroneous or outliers, we do not have sufficient evidence to do so; *e.g.*, multiple different vaccination sera samples have very low titers, which cause high error bars.²¹

We also investigated whether the anomalous behavior for SARS-2 described above could be responsible for the larger difference between the correlations computed over the training *vs.* test sets than was observed for influenza (compare panels C of Figs. 3 and 4 with panels C of Figs. 5 and 6), *i.e.* overfitting to the training set. To do this, we found the $\binom{8}{4}=70$ of the $\binom{9}{5}=126$ training/test partitions that contain the SARS-2 strain in the training set, and recomputed the average in the correlations using only those strains. Figures 5 and 6 (blue symbols; see figure caption) show that including the SARS-2 titers in the training data lowered the correlations in the training sets and increased correlations in the test set, which indicates reduced overfitting to training data. We interpret this finding as an additional indication that the IgG titers measured against the SARS-2 strain behave differently from the other titers (*e.g.*, that the sequence similarity is not the main titer determinant). It can also be interpreted as underscoring the importance of a diverse training set in reducing overfitting.

Finally, in Figs. 5 and 6 we colored the structure of the SARS-2 RBD by residue weights from the two models; the weights are also plotted as a function of residue position in Fig. S4B. Comparison of the figures shows that the

models predict similar sets of dominant residues. The residues that make the highest contributions to the antibody binding are not at the tip of the RBD, which contacts the angiotensin (ACE2) receptor prior to target cell entry, but near the ‘middle’ of the RBD. According to the X-ray and Cryo-EM structures of Barnes *et al.*²⁸, this part of the RBD contains epitopes which bind class 3 neutralizing antibodies, such as Regeneron REGN10987⁴² bnAb. Thus, the weights suggest that the vaccines elicit antibodies that target the more conserved parts of the RBD, in accord with the vaccine design.²¹ As was the case for influenza, epitope mapping experiments using competitive binding, would be needed to test the epitope predictions.

IV. DISCUSSION

Viral escape poses a major challenge for the development of vaccines for highly-mutable pathogens such as HIV, influenza and coronavirus, as rapidly emerging pathogen variants render host immunity raised by administered vaccines obsolete. Thus, it is essential to design vaccines that elicit broadly-reactive sera that can neutralize a wide variety of pathogen mutants, particularly, those that are not present in the vaccines. Toward this goal, we parametrized two models of reactivity of the sera obtained by vaccinating naïve mice with influenza or coronavirus nanoparticle vaccines.^{20,21} The models were able to recapitulate the experimental Ab titers to within experimental uncertainty, and produced Pearson and Spearman correlation coefficients in the range 0.6–0.9, depending on the data set used. Notably, the model parametrization procedure required only a modest amount of data (*e.g.*, 20 titer values used for parametrization/training, and 16–20 for testing). This indicates that using such models in the vaccine design process does not create an undue experimental burden. However, additional experimental data would improve model quality, *e.g.*, by reducing experimental uncertainty in the average titers (using more animals), by incorporating more strains into vaccine cocktails or testing panels, or by justifying the use of more complex models with additional parameters, such as multiple residue weights to represent different epitopes. In principle, sophisticated neural networks could be trained to model vaccine efficacy, such as those used to predict properties of small molecules,^{43,44} or antibody structures.⁴⁵ However, such models typically require very large amounts of data (thousands to millions of data points) to achieve high accuracy. These may be costly, even impossible, to obtain in vaccination experiments, and would probably not be transferable between vaccine types or vaccination protocols (a limitation also shared by our approach). Although the present models yield good results with few data points, they required significant human input, *e.g.*, choice of the functional form of the similarity function (Eq. (4)) or amino-acid encoding. For example, using the Atchley *et al.*³¹ encoding for the influenza vaccines yields a somewhat lower quality fit to the vaccine titers, with several model points outside of the experimental error bar (see supplementary Fig. S1B), and the distribution of residue weights was high for a portion of the hemagglutinin stem that is expected not to be exposed to antibodies in a trimer (Fig. S1D). For these reasons, and with the additional justification that the Grantham²⁹ encoding appears to be more grounded in the physical properties of amino acids, we chose the Grantham encoding, rather than the more empirical fit of Atchley *et al.*³¹.

The present models were developed to interpret vaccination data obtained from mosaic nanoparticle vaccines. The main reason for this choice is that such vaccines have been shown to direct the immune response to more conserved immunosubdominant epitopes,^{18,19,21,38} which are associated with eliciting bnAbs.^{40,46} The underlying reason for the greater breadth associated with such vaccines appears to be antibody avidity. Because antibodies and their membrane-bound counterparts – BCRs – have two fragments for antigen-binding (Fabs), and because antigen binding translates into a proliferation signal for the corresponding B cells during antibody evolution,⁴⁷ mosaic nanoparticle vaccines are expected to elicit antibodies directed to conserved parts on the antigen.¹⁸

Since Model 1 assumes that the maturing antibodies see an “average” strain, the encouraging results with Model 1 suggest that the vaccination sera contain bnAbs cross-reactive with multiple antigen strains. However, the fits to experimental titer averages produced with Model 2 are of similar quality, indicating that neither model is superior.

The ability to predict vaccine efficacy against a particular pathogenic variant is useful only if the variants have been identified. However, because the space of possible amino acid mutations is astronomical (even if not all 20 natural amino acids are allowed at all sequence positions), it is generally not possible to predict which specific strains will emerge in the future. Nevertheless, the effects of viral mutations can be partially characterized by high-throughput mutagenesis experiments⁷ to construct viral fitness landscapes, from which the most fit mutant sequences can be tested using models such as those presented here. Alternatively/additionally, one could use such models to maximize antibody breadth over a space of candidate vaccine designs (see *e.g.* Ref. 48) relative to an established panel of antigen variants.⁴⁹

Although the results described here are encouraging, it is important to discuss the limitations of the modeling approach. First, the model parametrizations described here are very unlikely to be directly transferable between different vaccine formulations because of a multitude of complex factors that determine the overall immune response, which are not explicitly represented in the models. Examples of such factors are nanoparticle *vs.* soluble ectodomain

TABLE III. Values of model parameters defined in *Methods*.

	α	β	τ	D	$n_{max}^\dagger$
Influenza					
Model 1	1	4	0.015–0.03	0.49	3000
Model 2	1	3	0.015–0.04	0.49	3000
Coronavirus					
Model 1	1	5	0.02–0.04	0.3	3000
Model 2	0.8	3.8	0.02–0.04	0.3	3000

[†]Number of gradient descent iterations in Eq. (7).

presentation, route of administration, choice of adjuvant, or choice of boosting protocol. Consequently, we expect the models to be somewhat general only within a particular class of vaccine formulation, *i.e.*, after the factors such those mentioned above are decided upon and fixed, so that the remaining differences between vaccines are dominated by antigen sequence variability. Even in these restricted scenarios, the models will likely need to be reparametrized. Fortunately, this is a simple task, particularly with the formalism of Eqs. (2)–(5), and does not require excessive preliminary experimentation. Further, if sufficient data are available for training, model sophistication can readily be improved. For example, an assumption that is implicitly made in our modeling is that a single weight array $\mathbf{W}$ can adequately describe the binding between the antibodies and the antigens. This assumption implies that all of the modeled vaccines are sufficiently similar so as to elicit antibodies to the same antigen epitopes, which is not true, in general. Thus, a trivial generalization of the present models is to include multiple weight arrays, *e.g.* one distance weight array per vaccine antigen in Eq. (5). Additionally, the overall similarity function (see Eq. (4)) can be composed of a sum of constituent functions, *e.g.*, for two weight arrays, $\mathbf{W}_1$ and $\mathbf{W}_2$,

$$f_{total} = f(d_1; \alpha_1, \beta_1) + f(d_2; \alpha_2, \beta_2), \quad (12)$$

where d_i is as in Eq. (3), with $\mathbf{W}_i$ in place of $\mathbf{W}$. The fitting/training of such models would be very similar to the procedure described here.

Finally, we do not explicitly take into account any preexisting immunity, or original antigenic sin (OAS), which is known to skew antibody responses toward previously encountered strains.⁵⁰ In principle, OAS could be incorporated in the model by including pre-vaccination titers to determine the direction of the skewness in antigen-experienced subjects.

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SUPPORTING MATERIAL

Supporting text containing sequence alignments of the antigenic sequences used here and supporting figures S1–S10 accompany this article, and can be accessed online free of charge. The computer code to reproduce the modeling data and figures in this paper, including the experimental data set used to train the models, can be found online at <https://github.com/ovchinnv/npvaccine-titer-model2022.git>.

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TABLE IV. Influenza strains used in the experiments²⁰ and in the present modeling.

ABBREVIATION	STRAIN NAME	ACCESSION #	V1	V2	V4	V8
C09	A/California/04/2009(H1N1)	ACS45035	✓	✓	✓	✓
V04	A/Viet Nam/1203/2004	AAW80717.1	–	–	✓	✓
J57	A/Japan/305/1957(H2N2)	AAA43185.1	–	–	–	✓
HK99	A/guinea fowl/Hong Kong/WF10/99	AAO46082.1	–	–	–	✓
AI68	A/Aichi/2/1968	AAA43178.1	–	✓	✓	✓
SH13	A/Shanghai/1/2013	EPI439486	–	–	✓	✓
JX13	A/Jiangxi/IPB13/2013(H10N8)	AHK10761	–	–	–	✓
HB09	A/swine/HuBei/06/2009(H4N1)	AFV33926	–	–	–	✓
P10	A/flat-faced bat/Peru/033/2010	AGX84934.1	–	–	–	–
WA79	A/wedge-tailed shearwater/Western Australia/2576/1979	ABB88138.1	–	–	–	–

V1, V2, V4, and V8 refer to the vaccination cocktails.

TABLE V. Coronavirus strains used in the experiments²¹ and in the present modeling.

NAME	ACCESSION #	HOST	V1	V4A	V4B	V8
SARS-2	MN985325.1	human	✓	✓	–	✓
RaTG13	QHR63300	bat	–	✓	–	✓
SHC014	KC881005	bat	–	✓	–	✓
Rs4081	KY417143	bat	–	✓	–	✓
Pang17	QIA48632	pangolin	–	–	✓	✓
RmYN02	EPI_ISL_412977	bat	–	–	✓	✓
Rf1	DQ412042	bat	–	–	✓	✓
WIV1	KF367457	bat	–	–	✓	✓
SARS	AAP13441.1	human	–	–	–	–
Yun11	JX993988	bat	–	–	–	–
BM-4831	NC_014470	bat	–	–	–	–
BtKY72	KY352407	bat	–	–	–	–

V1, V4a, V4b, and V8 refer to the vaccination cocktails.

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